Register

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Abstract

A CTCPU register holds the complete record of one number. It consists of eight rows, each with eight cells. Since a cell is three bits, the register is 192 bits. Each row has a separate job: the number's octal digits, four shadow rows, rough cache, scale and type. These 192 bits are not all the number itself. The number is only in the eight digits of the first row. The other seven rows hold other representations of the same number and information about it.

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1. Capital R and small r

There are eight registers in the core: $R0...R7$. Operations use them, for example `ADD R3, R5`. Each large register contains eight rows. I give these small r names: rm, $rs1...rs4$, rc, $rpow$, $rtyp$. $R3$ has its own rm and its own $rs1$. $rs1$ is not $R1$.

2. Rows

Row	Name	Contents
r0	rm	**the number's eight octal digits, $h_7...h_0$**
r1	$rs1$	**shadow, first level: seven exponent cells, a cell empty**
r2	$rs2$	**shadow, second level: seven exponents, digit a cut on descent**
r3	$rs3$	**shadow, third level: seven exponents, a**
r4	$rs4$	**shadow, fourth level: seven exponents, a**
r5	rc	**rough cache, eight digits**
r6	$rpow$	**scale declaration: value $= rm \cdot 8^{-rpow}$**
r7	$rtyp$	**type: cell 0 is the type code, the other seven cells are open**

In the shadow rows, the cells from left to right are the exponents of 2, 3, 5, 7, 11, 13 and 17. The last cell is a. In $rs1$, the a cell is empty, because there is no descent from rm to

$rs1$. The chain starts at $rs1$. Three descents leave three filled a cells.

3. An example: 2124

I write the number $2124 = 4114_8 = 2^2 \cdot 3^2 \cdot 59$ into a register. Resolver's conversion is Example 4 in the gear model.

R3: N = 2124 = 4114_8

rm	0	0	0	0	4	1	1	4	eight number digits
rs1	2	2	0	0	0	0	0	·	level 1 exponents | a
rs2	0	0	0	1	0	0	0	3	level 2 exponents | a
rs3	0	0	0	0	0	0	0	·	level 3
rs4	0	0	0	0	0	0	0	·	level 4
rc	0	0	0	0	0	0	0	1	rough cache
rpow	0	0	0	0	0	0	0	0	scale
rtyp	0	·	·	·	·	·	·	·	type | open

rs row channel order: 2 3 5 7 11 13 17 | a
8 rows × 8 cells × 3 bits = 192 bits
2124 = 2^2 · 3^2 · (8 · 7 + 3)

Figure 1. 2124 in register R3. Thick lines separate the number, shadow and other fields.

The rm row holds the number itself, 00004114. In the $rs1$ row, the exponents of 2 and 3 are 2, meaning $2^2 \cdot 3^2 = 36$ has been stripped at the first level. In the $rs2$ row, the exponent of 7 is 1 and $a = 3$: the last digit of 59 has been cut and 7 has descended to the second level, $59 = 8 \cdot 7 + 3$. The rc row holds the final residual 1. The initial rough factor 59 is reconstructed from the lower level. $rpow$ is 0, because the number is an integer. Cell 0 of $rtyp$ is also 0, meaning the type is a positive integer.

Reading the shadow rows upwards reconstructs the number:

$$2124 = 2^2 \cdot 3^2 \cdot (8 \cdot 7 + 3)$$

4. Rules

If all eight cells of rm are 0, no shadow is built, because zero has no factors. When rc holds the first rough factor, $rc \neq 1$ says that the number has a rough part and removes the need for a separate rough flag. When rc holds the final residual, the lower ladder levels must also be read to determine the initial rough factor. rc is full width, because a number can be entirely rough. For example, $77777775_8 = 16\,777\,213$ is prime and its rough part is itself.

$rpow$ **exists to declare the number's precision. With** rm **fixed, increasing** $rpow$ **makes the value** $rm \cdot 8^{-rpow}$ **smaller. The value is only scaled down. There is no separate status field. Statuses and type are in** $rtyp$**. Details are in Types.**

5. Open questions

Exactly how the eight cells of the $rpow$ **row will be used and the layout of the seven cells of** $rtyp$ **other than cell 0 have not yet been decided. Whether** rc **holds the entire first rough factor or the last unprocessed residual is also open. These two meanings cannot replace one another when reading back. The need for a separate ROUGH status depends on this choice.**

There is another use for the shadow rows. In the gear model, the same four rows are used as the octal digits of a single exponent, and an exponent can reach 4095. In the layout described here, each row is a level and each cell holds an exponent from 0 to 7. Which of the two uses will be settled on is open.

Related pages: Cell, Core and Parts.

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